









Modelling Financial Markets

Chapter 4: Portfolio Optimisation



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Why Does Estimating the Optimum Change the Problem?

Easy optimum, hard estimate

- A **portfolio** is a weight vector $\mathrm{w} = (w_1, \ldots, w_N)^\top$ with $\mathrm{w}^\top 1 = 1$; optimisation chooses w to maximise a criterion under constraints
- The optimum is easy to write with known means and covariances, hard to estimate: the optimiser loads on the largest estimation errors
  - 16 assets: 16 means + 136 variances and covariances = 152 parameters, estimated from 60 monthly observations
- Can any estimated rule beat equal weights out of sample, after costs?

What the Chapter Contains

Core path

- Mean–variance: global minimum-variance (GMV) and tangency weights, the frontier, constraints
- Estimation risk: why in-sample Sharpe ratios are too optimistic
- Three repairs of the inputs: covariance shrinkage, constraints, Black–Litterman
- Rules that need only Σ: equal risk contribution (ERC) and Hierarchical Risk Parity (HRP)
- Out-of-sample evaluation: turnover and costs, Sharpe-difference tests, Holm and Benjamini–Hochberg
- The application on the Bucharest Stock Exchange (BVB)

Deeper study

- Trading costs of a large investor (case study)
- Research assisted by artificial intelligence (AI) and data snooping
- Appendix: derivations and complete tables
- Textbook: Franke, Härdle & Hafner (2019), Part I

What you should be able to do

- compute minimum-variance, tangency and equal-risk weights
- explain why estimation error favours 1/N
- test a Sharpe difference after costs and multiple testing

Data, Assets and the 1/N Benchmark

Universe	Assets	Returns	Currency
Sectors	9 US sector ETFs	1999/01 – 2026/07	USD
Multi-asset	8 ETFs: equities (3), Treasuries (2), credit (2), gold	2007/05 – 2026/07	USD
Combined	9 sectors + the 7 multi-asset ETFs other than SPY (S&P 500) (16 ETFs)	2007/05 – 2026/07	USD
Romania	8 BVB blue chips; the BET and BET-TR indices	2014/10 – 2026/08	RON

- ETF (exchange-traded fund): a fund traded on an exchange that tracks an index
- Monthly simple returns from month-end prices on common trading days
  - ETFs and stocks with dividends reinvested
  - BET (Bucharest Exchange Trading), the BVB benchmark, is a price index; BET-TR (total return) reinvests dividends
- Excess returns over the one-month Treasury bill (Kenneth French Data Library); BVB: $R_f = 0$
- **1/N**: equal weights $w_i = 1/N$, rebalanced every month; no estimation at all, the benchmark of every comparison

The Assets in Detail

- **Sectors** (9 US sector ETFs)
  - XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology
  - XLP staples, XLU utilities, XLV health care, XLY discretionary
- **Multi-asset**
  - equities: SPY (S&P 500), EFA (developed markets outside the US and Canada), EEM (emerging markets)
  - bonds: TLT (US Treasuries, 20+ years), IEF (US Treasuries, 7–10 years), LQD (investment grade), HYG (high yield); gold: GLD
- **BVB** (8 blue chips)
  - TLV Banca Transilvania, SNP OMV Petrom, BRD BRD–Groupe Société Générale, TGN Transgaz
  - SNG Romgaz, SNN Nuclearelectrica, EL Electrica, TEL Transelectrica

Notation (1/2)

- **Returns**
  - simple return $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$; $P_{i,t}$: the month-end price of asset i
  - excess return $R^e_{i,t} = R_{i,t} - R_{f,t}$ (in the seminar: r_t and r^e_t); $R_{f,t}$: the risk-free rate
  - log return $\ln(P_{i,t}/P_{i,t-1}) = \ln(1 + R_{i,t})$: adds up over time, not across assets; e.g. a price from 100 to 110: $R = 10\%$, $\ln 1.1 = 9.53\%$
  - portfolios use simple returns, because $R_{p,t} = \sum_i w_i R_{i,t}$ holds exactly
- **Portfolio** of N assets
  - w: vector of weights; $\boldsymbol{\mu}$: vector of expected (excess) returns; 1: vector of ones
  - $\boldsymbol{\Sigma}$: the $N \times N$ covariance matrix of the returns
  - portfolio mean $\mathrm{w}^\top \boldsymbol{\mu}$, portfolio variance $\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$

Notation (2/2)

- **Sharpe ratio** (Sharpe (1966)): $SR_p = \mathbb{E}[R_p^e]/\sigma(R_p^e)$
  - the mean excess return per unit of volatility σ
  - monthly figures annualised with $\times 12$ (mean) and $\times\sqrt{12}$ (volatility, Sharpe)
- **Maximum drawdown**: $\min_t(W_t/\max_{s\le t} W_s - 1)$
  - the largest fall of wealth W_t from a previous peak
  - example: wealth rises to 1.50, then falls to 1.20: $1.20/1.50 - 1 = -20\%$
- **Out of sample** (OOS): the weights held in month t use only data up to month $t-1$
- **Rules of this chapter**
  - MV (mean–variance): tangency portfolio from sample moments; GMV: global minimum variance
  - MV-LO and GMV-LO: their long-only (LO) versions; GMV-LW: GMV with Ledoit–Wolf (LW) covariance shrinkage
  - ERC (equal risk contribution) and HRP: risk-based rules; 1/N: equal weights

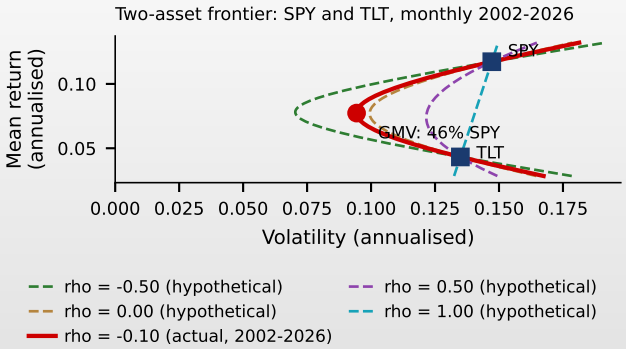
The Markowitz Problem

- Markowitz (1952): investors care about the mean and the variance of portfolio returns
- **Minimum-variance frontier**: minimum variance for every target mean m
  - its upper branch, $m \geq \mu_{\text{GMV}}$ (the mean of the GMV portfolio), is the **efficient frontier**

$$\min_{\mathrm{w}} \ \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w} \quad \text{s.t.} \quad \mathrm{w}^\top \boldsymbol{\mu} = m, \ \ \mathrm{w}^\top 1 = 1$$

  - Objective: portfolio variance; constraints: the target mean m and full investment (weights sum to 1)
- Utility form, risk aversion $\gamma > 0$: $\max_{\mathrm{w}} \ \mathrm{w}^\top \boldsymbol{\mu} - \frac{\gamma}{2} \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$ s.t. $\mathrm{w}^\top 1 = 1$
  - as γ varies, the solution traces exactly the efficient branch
- Weights may be negative: a **short sale** (selling a borrowed asset)

Two Assets: How Correlation Diversifies



- Two assets, correlation ρ: $\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$; w: weight of asset 1, σ_1, σ_2: volatilities, σ_p: portfolio volatility
- Minimised at $w^* = \dfrac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- SPY and TLT, monthly 2002–2026: $\rho = -0.10$, $w^* = 46\%$ SPY, volatility 9.4% against SPY 14.7% and TLT 13.5%
- Chart: means of total returns and annualised volatilities, in decimals (0.10 = 10%)
- The lower ρ, the more the curve bends to the left; at $\rho = 1$ the long-only portfolios shown have no diversification benefit
- Time-varying correlation: Chapter 6

MFM_ch4_two_asset

GMV and Tangency Portfolios for N Assets

- GMV: $\min_{\mathrm{w}} \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$ s.t. $\mathrm{w}^\top 1 = 1$; Lagrangian $\mathcal{L} = \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w} - \lambda(\mathrm{w}^\top 1 - 1)$, $\boldsymbol{\Sigma}$ positive definite
  - λ: Lagrange multiplier of the budget constraint; first-order condition $2\boldsymbol{\Sigma}\mathrm{w} = \lambda 1$, so $\mathrm{w} = \frac{\lambda}{2}\boldsymbol{\Sigma}^{-1}1$; the constraint gives $\frac{\lambda}{2} = 1/A$, $A = 1^\top \boldsymbol{\Sigma}^{-1} 1$
- Tangency: maximise $\mathrm{w}^\top \boldsymbol{\mu} / \sqrt{\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}}$, $\boldsymbol{\mu}$ the expected **excess** returns; the condition $\boldsymbol{\mu} \propto \boldsymbol{\Sigma}\mathrm{w}$ gives $\mathrm{w} \propto \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ ($\propto$: proportional to); θ: the maximum Sharpe ratio

	GMV	**Tangency (MV)**
Weights	$\mathrm{w}_{\mathrm{GMV}} = \boldsymbol{\Sigma}^{-1}1/A$	$\mathrm{w}_{\mathrm{MV}} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}/B$, $B = 1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$
Result	variance $\sigma^2_{\mathrm{GMV}} = 1/A$	squared Sharpe $\theta^2 = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$
Needs	$\boldsymbol{\Sigma}$ only	$\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$; Sharpe $\mathrm{sign}(B)\,\theta$

- The normalised $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}/B$ is the maximum only if $B > 0$; $B < 0$ gives the minimum-Sharpe portfolio, $B = 0$ no fully invested solution

Separation and the Frontier in Closed Form



James Tobin, Nobel Prize in Economics 1981

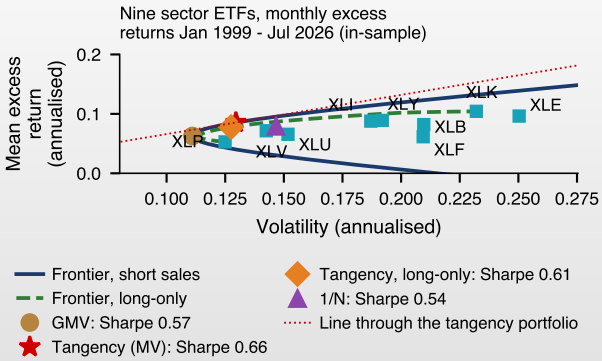
Photo: unknown author (1962); public domain; Wikimedia Commons

- Tobin (1958): add a riskless asset paying R_f
  - every investor holds the **same** risky portfolio, the tangency portfolio, mixed with cash or borrowing
  - risk aversion changes only the mix: **two-fund separation**, the bridge to the Capital Asset Pricing Model (CAPM; Chapter 3)
- Merton (1972): with short sales, the frontier is a hyperbola; Σ positive definite, μ not proportional to 1
  - A, B as on the previous slide; $C = \mu^\top \Sigma^{-1} \mu = \theta^2$, $D = AC - B^2 > 0$

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$

- $\sigma^2(m)$: the smallest variance attainable with mean m; a parabola in the (σ^2, m) plane, the hyperbola above in the (σ, m) plane (derivation: ▸ Appendix)
  - the GMV sits at $m = B/A$, $\sigma^2 = 1/A$
  - every frontier portfolio mixes any two distinct frontier portfolios, e.g. GMV and tangency (if $B \neq 0$)

The Frontier of US Sectors



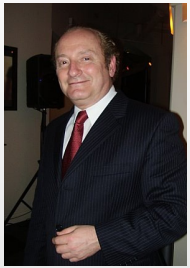
- Monthly excess returns, January 1999 – July 2026 ($T = 331$), axes annualised, in decimals; all parameters estimated on the full sample: an **in-sample** picture
- Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); 1/N: 0.54
- The tangency weights include short positions: -0.37 financials, -0.21 materials
- The largest long position of the tangency portfolio: $+0.37$ health care
- Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero, a **corner solution**
- The gain over 1/N is small even with perfect hindsight

MFM_ch4_frontier

Constraints

- **Long-only** (LO): $w_i \geq 0$; no closed form, solved by quadratic programming
- Position limits: $w_i \leq 0.2$; sector or asset-class limits: $\sum_{i \in G} w_i \leq c_G$, G a group of assets, c_G its cap
- Leverage limit: $\sum_i |w_i| \leq L$; $\sum_i |w_i|$: gross exposure, $L = 1$ means no short sales
- Frost & Savarino (1988): constraining weights improved out-of-sample performance; the reason: constraints act as shrinkage of the covariance matrix (Jagannathan & Ma (2003); slide Constraints Are Shrinkage Too)

Means Are Hard to Estimate: Merton's Argument

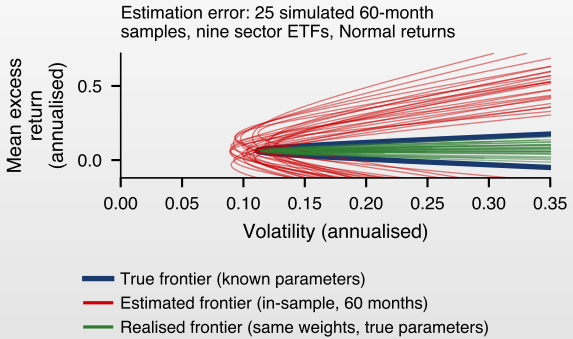


Robert C. Merton, Nobel Prize in Economics 1997

Photo: Digarnick (2006); public domain; Wikimedia Commons

- Merton (1980): Y years sampled h times a year, i.i.d. Normal returns (a diffusion) with annual mean μ and variance σ^2
  - $\mathrm{Var}(\hat{\mu}) = \sigma^2/Y$: independent of h; only the **span** of the sample helps
  - $\mathrm{Var}(\hat{\sigma}^2) \approx 2\sigma^4/(hY) \to 0$ as $h \to \infty$ (Normal; in general $(\kappa - 1)\sigma^4/(hY)$, κ the kurtosis): variances sharpen with frequency
- SPY, $Y = 24$, $\sigma = 14.7\%$: standard error 3.0%, 95% interval $[5.8\%, 17.6\%]$ for a mean of 11.7%
- A standard error of 1% needs $(0.147/0.01)^2 \approx 217$ years
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances; rules without means (GMV, risk parity) are more robust

Error Maximisation: Estimated versus Realised Frontiers



- Michaud (1989): optimisers are **estimation-error maximisers**; Best & Grauer (1991): a tiny change in one mean moves many weights
- True parameters = sector moments 1999–2026; draw 60 Normal months, optimise, evaluate with the true parameters
- Chart: 25 simulated 60-month samples; red: estimated frontiers, too optimistic; green: the same portfolios under the true parameters, all below the true frontier
- 2000 simulated tangency portfolios: median estimated maximum Sharpe 1.56, median true Sharpe 0.30, true optimum 0.66
- 96% have a true Sharpe below 1/N (0.54), 7.5% below zero
- In 11% of samples $\hat{B} = 1^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}} < 0$, so the tangency formula returns the minimum-Sharpe portfolio

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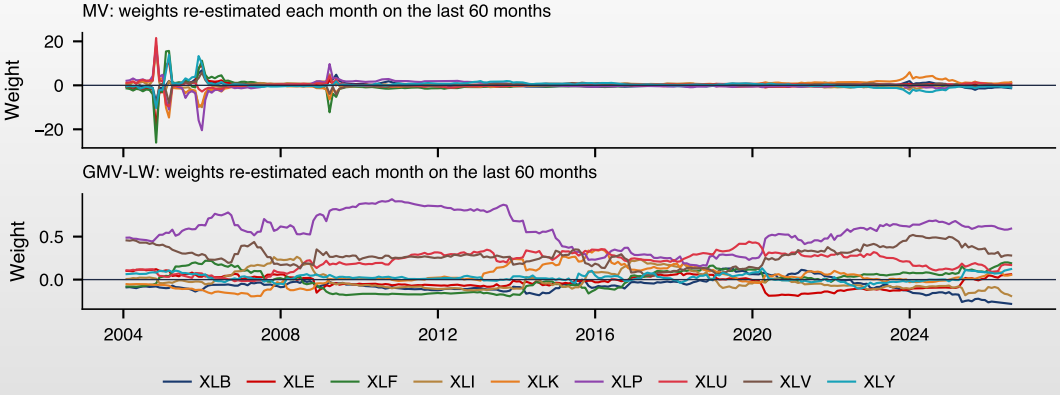
The Plug-in Sharpe Ratio Is Biased

- i.i.d. $N(\mu, \Sigma)$ returns, T months, N assets (Kan & Zhou (2007); Wishart moments in the Appendix)
  - $\hat{\theta}^2 = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}$: the in-sample squared maximum Sharpe ratio, from the maximum-likelihood $\hat{\mu}$, $\hat{\Sigma}$; θ^2: its true value

$$\mathbb{E}[\hat{\theta}^2] = \frac{T}{T-N-2}\left(\theta^2 + \frac{N}{T}\right) = \frac{T\theta^2 + N}{T-N-2}, \qquad T > N+2$$

- Sectors: $N = 9$, $T = 60$, true $\theta = 0.66$ a year: $\sqrt{12\,\mathbb{E}[\hat{\theta}^2]} = 1.65$ a year, a root mean square, not a median
  - the simulated median 1.56 is lower: $\hat{\theta}$ is right-skewed, and the simulation uses the divisor $T-1$ in $\hat{\Sigma}$; derivation: Seminar 4, A8
- Unbiased estimator of θ^2: $\hat{\theta}^2_u = \left[(T-N-2)\,\hat{\theta}^2 - N\right]/T$
  - It can be negative; truncating at zero gives a nonnegative but upward-biased estimator, and $\sqrt{\hat{\theta}^2_u}$ is not unbiased for θ
- Sectors, full sample ($T = 331$): plug-in $\hat{\theta} = 0.66$, bias-corrected $\sqrt{\hat{\theta}^2_u} = 0.31$ a year
  - more than half of the in-sample Sharpe is noise, even with 27 years
- The tangency weights are the ordinary least squares (OLS) coefficients of a regression of 1 on the returns (Britten-Jones (1999); Appendix; Seminar 4, A9 and B6)

Unstable Weights (1/2)



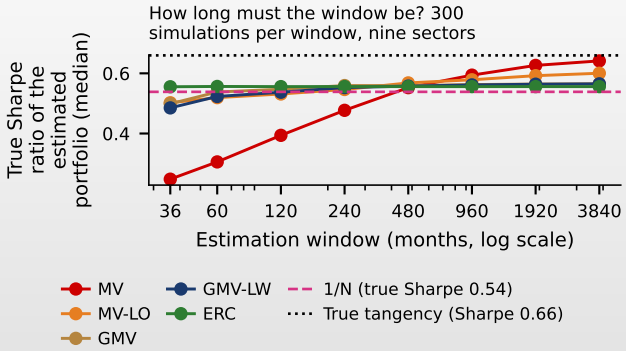
MFM_ch4_oos_backtest

Unstable Weights (2/2)

- Nine sector ETFs; each month the weights use only the previous 60 months
- MV: median gross exposure $\sum_i |w_i| = 4.9$, above 3 in 90% of the months; largest single weight 26 in absolute value
- Explosions in 2004–2006 and 2009: the denominator $1^\top \hat{\Sigma}^{-1} \hat{\mu}$ is close to zero, so the tangency weights have very heavy tails
  - Negative in 12 of 271 windows: MV then holds the minimum-Sharpe portfolio
- GMV-LW (bottom): gross exposure 1.5, largest weight 0.94
- Unstable weights mean high turnover, hence high trading costs

MFM_ch4_oos_backtest

How Long Must the Window Be to Beat 1/N?



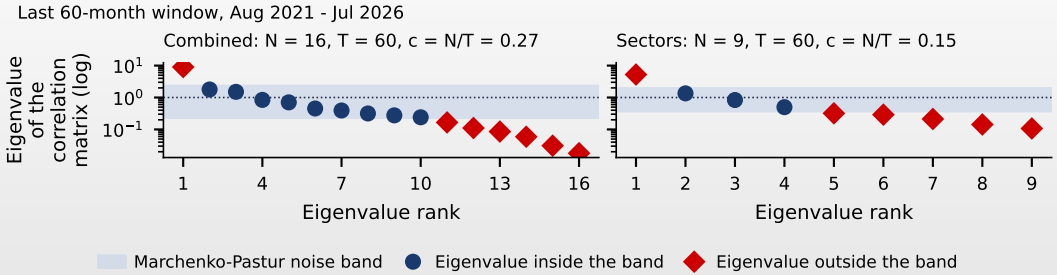
- DeMiguel, Garlappi & Uppal (2009): 14 models on 7 data sets; none beats 1/N consistently in Sharpe ratio, certainty-equivalent return (mean $-\frac{\gamma}{2}\times$ variance, the riskless return worth the same utility) or turnover; their calibration needs about 3000 months for 25 assets
- Here: 300 simulated samples per window, population parameters = sector moments
- MV beats 1/N only with 480 months (40 years): median true Sharpe 0.55 vs 0.54; with 60 months 0.31 (0.30 in the 2000 simulations of the error-maximisation slide)
- MV-LO, GMV, GMV-LW are close to 1/N from 60 months on; ERC is almost insensitive, 0.56
- In this calibration 1/N is nearly efficient (true tangency 0.66): simulation evidence, not an empirical test

MFM_ch4_window_length

Why the Sample Covariance Matrix Fails

- Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- **Condition number**: largest over smallest eigenvalue; a large value means $\boldsymbol{\Sigma}^{-1}$ amplifies noise
- Optimisers invert $\boldsymbol{\Sigma}$, so they load on the directions with the smallest estimated variance
- The demeaned sample matrix has rank at most $T - 1$: singular, not invertible, whenever $T \leq N$
- **Marchenko–Pastur** benchmark (Marčenko & Pastur (1967)): i.i.d. returns with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ (I: identity matrix)
  - as $N, T \to \infty$ with $N/T \to c \in (0, 1)$, the sample eigenvalues fill
    $$\left[\lambda_-, \lambda_+\right] = \left[\sigma^2(1 - \sqrt{c})^2,\ \sigma^2(1 + \sqrt{c})^2\right]$$
  - λ_-, λ_+: the smallest and largest eigenvalue that pure noise produces
  - $c = 16/60 = 0.27$: band $[0.23, 2.30]$ for a correlation matrix ($\sigma^2 = 1$), a ten-fold spread from noise alone
  - Laloux et al. (1999): eigenvalues outside the band are candidates for structure; those inside cannot be told from noise by this comparison alone

Sample Eigenvalues versus the Noise Band



- Correlation matrices of monthly excess returns, last 60-month window (August 2021 – July 2026); shaded band: $[\lambda_-, \lambda_+]$ for $\sigma^2 = 1$
- Only the market mode lies above λ_+: 9.0 (56% of the trace) for 16 ETFs, 5.2 for nine sectors ($\lambda_+ = 1.92$); with the noise variance rescaled to $\sigma^2 = 1 - \lambda_1/N$ (Laloux et al. (1999)), three eigenvalues exceed the band in each universe
- Six of 16 eigenvalues lie **below** the band: near-riskless combinations, such as long TLT and short IEF
- GMV puts large weights on these combinations; if their variance is underestimated, so is the GMV risk

MFM_ch4_inference

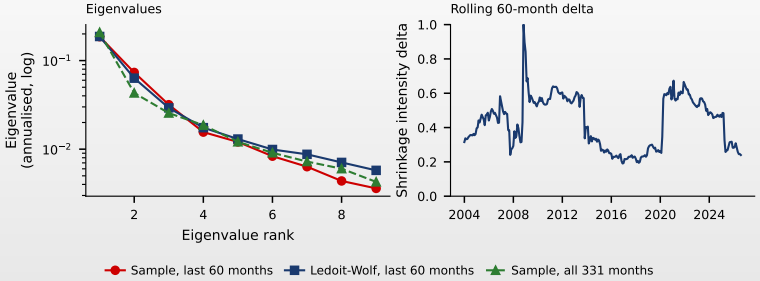
The Shrinkage Estimator

- Shrink the sample matrix S towards a structured target F

$$\hat{\Sigma}_{\mathrm{LW}} = \delta\,\mathsf{F} + (1-\delta)\,\mathsf{S}, \qquad 0 \le \delta \le 1$$

- Targets
  - Single-index (market) model: Ledoit & Wolf (2003)
  - Scaled identity: Ledoit & Wolf (2004a)
  - Constant correlation, $f_{ii} = s_{ii}$ and $f_{ij} = \bar{r}\sqrt{s_{ii}s_{jj}}$ for $i \neq j$, $\bar{r}$ the average sample correlation: Ledoit & Wolf (2004b), used here
- δ trades the variance of S against the bias of F; δ is large when S is noisy or the target F is close to $\boldsymbol{\Sigma}$; long samples, $\delta \to 0$ (the optimal δ^* in the Appendix)

Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026: $\delta = 0.24$; the smallest eigenvalues are pulled up, the largest almost unchanged; condition number 51.6 $\rightarrow$ 32.5
- Rolling δ: mean 0.43, range 0.19–1.00; $\delta = 1$ in the early windows, where 60 months are too few to beat the target
- GMV-LW: the GMV portfolio computed with $\hat{\boldsymbol{\Sigma}}_{\mathbf{LW}}$

MFM_ch4_shrinkage

Constraints Are Shrinkage Too

- Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix

$$\tilde{S} = S - (\boldsymbol{\lambda} 1^\top + 1\boldsymbol{\lambda}^\top), \qquad \mathcal{L} = \tfrac{1}{2} w^\top S w - \lambda_0 (w^\top 1 - 1) - \boldsymbol{\lambda}^\top w$$

  - $\boldsymbol{\lambda} \geq 0$: the Lagrange multipliers of the constraints $w_i \geq 0$ in $\mathcal{L}$; $\lambda_i > 0$ only where the constraint binds
  - The first-order condition $Sw = \lambda_0 1 + \boldsymbol{\lambda}$ and $\boldsymbol{\lambda}^\top w = 0$ give $\tilde{S}w = \lambda_0 1$: the GMV condition for $\tilde{S}$
  - step by step, with the budget $1^\top w = 1$: $\tilde{S}w = Sw - \boldsymbol{\lambda}(1^\top w) - 1(\boldsymbol{\lambda}^\top w) = \lambda_0 1 + \boldsymbol{\lambda} - \boldsymbol{\lambda} - 0$
- $\tilde{s}_{ij} = s_{ij} - \lambda_i - \lambda_j$: a binding constraint on asset i lowers its covariances, which shrinks the large estimates
  - why there: the unconstrained GMV shorts assets with high covariances with the others, so $w_i \geq 0$ binds on exactly those assets, whose large covariance estimates are the most likely to be inflated by noise
- Imposing the "wrong" constraint (no short sales) can reduce out-of-sample risk; the effect is largest when covariances come from few observations

Constraints as Shrinkage: A Two-Asset Example

- $\sigma_1 = 25\%$, $\sigma_2 = 10\%$, $\rho = 0.5$: $s_{11} = 0.0625$, $s_{22} = 0.0100$, $s_{12} = 0.0125$
- Unconstrained GMV: $w_1^* = (s_{22} - s_{12})/(s_{11} + s_{22} - 2s_{12}) = -0.0025/0.0475 = -0.053$
  - the volatile asset is shorted because it co-moves strongly with the other; volatility 9.93%
- Long-only: $w_1 \geq 0$ binds, w $= (0, 1)$, volatility 10%
  - first-order condition: Sw $= (s_{12}, s_{22}) = (0.0125, 0.0100) = \lambda_0 1 + \boldsymbol{\lambda}$, so $\lambda_0 = 0.0100$, $\boldsymbol{\lambda} = (0.0025, 0)$
- Modified matrix: $\tilde{s}_{12} = 0.0125 - 0.0025 = 0.0100$, $\tilde{s}_{11} = 0.0625 - 2(0.0025) = 0.0575$, $\tilde{s}_{22} = 0.0100$
  - correlation lowered from 0.5 to $0.0100/\sqrt{0.0575 \times 0.0100} \approx 0.42$; the variance of asset 1 also falls
  - unconstrained GMV of $\tilde{\mathrm{S}}$: $w_1^* = (0.0100 - 0.0100)/(\cdots) = 0$, the long-only solution

Nonlinear Shrinkage

- Linear shrinkage mixes two matrices with one δ; **nonlinear** shrinkage keeps the sample eigenvectors and corrects each eigenvalue separately
  - Ledoit & Wolf (2017): tuned for portfolio selection; Ledoit & Wolf (2020): an analytical formula, no optimisation
- Out of sample, 60-month window, GMV volatility
  - Sectors: sample 11.90%, linear (constant correlation) 11.80%, nonlinear 11.62%
  - 16 ETFs: sample 4.06%, linear 4.38%, nonlinear 4.31%: better than the linear target, not than the raw matrix
- A constant correlation fits neither stock–stock (positive) nor stock–bond (negative) pairs: the target matters (Seminar 4, B7)
- The raw sample matrix becomes unusable as N approaches T (singular for $T \leq N$); at $N = 16$, $T = 60$ it was still competitive

From Covariance Shrinkage to Mean Shrinkage



Fischer Black (1938–1995)

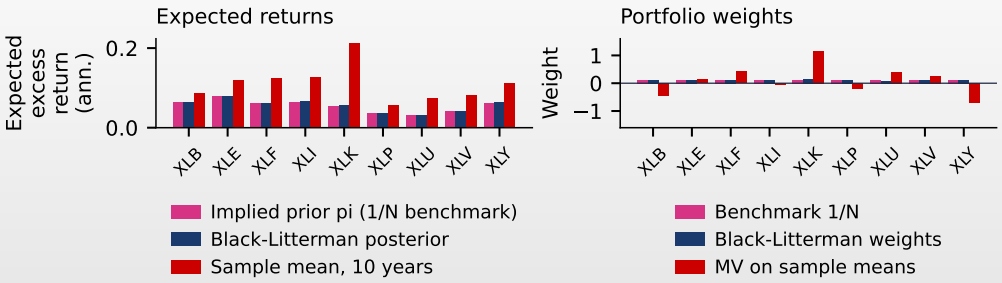
Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons

- Shrinkage fixes $\hat{\boldsymbol{\Sigma}}$; Black & Litterman (1992), developed at Goldman Sachs, shrinks the means towards those implied by a **benchmark** portfolio w_b
- Reverse optimisation: a mean–variance investor with risk aversion $\delta_{\mathbf{RA}}$ and access to cash holds $w^* = (\delta_{\mathbf{RA}} \boldsymbol{\Sigma})^{-1} \boldsymbol{\mu}$ in risky assets; the returns that make w_b optimal are

$$\boldsymbol{\pi} = \delta_{\mathbf{RA}} \boldsymbol{\Sigma} w_b$$

- $\boldsymbol{\pi}$: **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- Here: no market capitalisations for the sectors, so $w_b = 1/N$, $\delta_{\mathbf{RA}} = 2.5$
- Then add the investor's **views**, weighted by their confidence: the weights move away from the benchmark only where views exist

Black–Litterman on US Sectors



- Σ from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year; implied XLK minus XLU spread 2.2%, posterior 2.6%
- Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged; the view is relative (P1 = 0), so the tilt $\propto P^\top$ sums to zero and the portfolio stays fully invested
- Sample means instead: XLK 21.3% a year; MV weights +115% XLK, −68% XLY, −46% XLB
- The XLK and XLU weights by hand: Seminar 4, A7

MFM_ch4_black_litterman

Views and the Posterior

- K views: $\mathrm{P}\boldsymbol{\mu} = \mathrm{q} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N(0, \boldsymbol{\Omega})$; prior $\boldsymbol{\mu} \sim N(\boldsymbol{\pi}, \tau\boldsymbol{\Sigma})$
  - P: $K \times N$ matrix, row k = the portfolio of view k; q: the view returns; $\boldsymbol{\Omega}$: their uncertainty; τ: scales the uncertainty of the prior $\boldsymbol{\pi}$
- Posterior mean: a precision-weighted average; weights $\mathrm{w}_{\mathbf{BL}} = (\delta_{\mathbf{RA}}\boldsymbol{\Sigma})^{-1}\boldsymbol{\mu}_{\mathbf{BL}}$

$$\boldsymbol{\mu}_{\mathbf{BL}} = \left[(\tau\boldsymbol{\Sigma})^{-1} + \mathrm{P}^{\top}\boldsymbol{\Omega}^{-1}\mathrm{P}\right]^{-1}\left[(\tau\boldsymbol{\Sigma})^{-1}\boldsymbol{\pi} + \mathrm{P}^{\top}\boldsymbol{\Omega}^{-1}\mathrm{q}\right]$$

- Limits: $\boldsymbol{\Omega} \to 0$ gives $\mathrm{P}\boldsymbol{\mu}_{\mathbf{BL}} = \mathrm{q}$ exactly; $\boldsymbol{\Omega} \to \infty$ gives $\boldsymbol{\pi}$; without views, the benchmark itself
- He & Litterman (2002): $\tau = 0.05$, $\boldsymbol{\Omega} = \mathrm{diag}(\mathrm{P}\,\tau\boldsymbol{\Sigma}\,\mathrm{P}^{\top})$; with $\boldsymbol{\Sigma}$ in the weights, only the assets in the views move
- One asset, $\pi = 5\%$, $\tau\sigma^{2} = 0.00128$, view 8% with variance 0.0004: $\mu_{\mathbf{BL}} = 7.29\%$, view weight 0.76
- The same estimator is a mixed regression estimated by generalised least squares (GLS) (Theil & Goldberger (1961); Appendix; Seminar 4, A6)

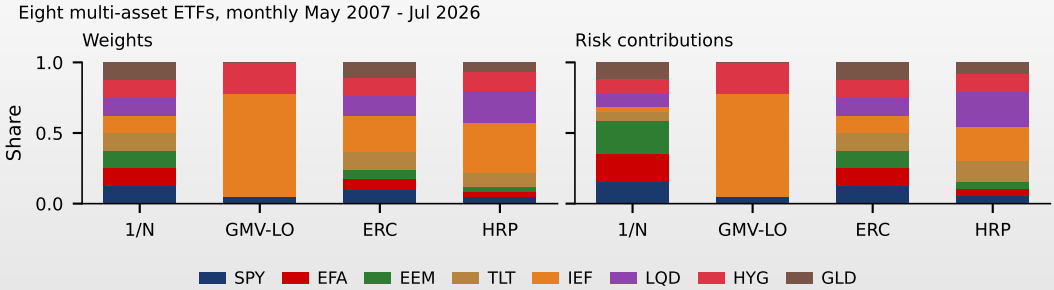
Black–Litterman: What δ_{RA} and τ Do

- $\delta_{\text{RA}} = 2.5$ and $\tau = 0.05$ are the values of He & Litterman (2002)
- **Risk aversion**: premultiply $\boldsymbol{\pi} = \delta_{\text{RA}} \boldsymbol{\Sigma} \mathrm{w}_b$ by $\mathrm{w}_b^\top$

$$\delta_{\text{RA}} = \frac{\mathrm{w}_b^\top \boldsymbol{\pi}}{\mathrm{w}_b^\top \boldsymbol{\Sigma} \mathrm{w}_b} = \frac{\text{benchmark excess return}}{\text{benchmark variance}}$$

  - e.g. 6% excess return at 15.5% volatility: $0.06/0.155^2 = 2.5$
  - $\boldsymbol{\pi}$ is linear in δ_{RA}: doubling it doubles the implied XLK minus XLU spread of 2.2%
- **Prior uncertainty**: $\tau\boldsymbol{\Sigma}$ is the uncertainty about the mean; one reading is $\tau \approx 1/Y$, the variance of a mean estimated over Y years relative to $\boldsymbol{\Sigma}$ (Merton's argument), so $\tau = 0.05$ is about 20 years of data
- With $\boldsymbol{\Omega} = \mathrm{diag}(\mathrm{P}\tau\boldsymbol{\Sigma}\mathrm{P}^\top)$ and one view, τ cancels: the posterior spread is the midpoint $(2.2\% + 3\%)/2 = 2.6\%$, and the weights do not depend on τ
- The confidence in the view is therefore set by $\boldsymbol{\Omega}$ relative to $\tau\boldsymbol{\Sigma}$, not by τ alone

Weights versus Risk Contributions



- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ($T = 231$); chart shares in decimals ($0.5 = 50\%$)
- 1/N: equities and high yield receive half of the weight but 69% of the risk; ERC: 12.5% of the risk from every asset, IEF gets the largest weight (25%); long-only GMV: 73% IEF, 22% HYG, 5% SPY, five assets at zero
- Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%: ERC lies between 1/N and GMV, as the theory predicts

MFM_ch4_risk_contrib

Risk Contributions

- Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^{N} w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^{N} \frac{w_i (\Sigma \mathrm{w})_i}{\sigma_p}$$

- **Risk contribution** of asset i: $RC_i = w_i(\Sigma \mathrm{w})_i/\sigma_p$; percentage share $RC_i/\sigma_p = w_i(\Sigma \mathrm{w})_i/\sigma_p^2$
- Worked example, 50/50 in SPY and IEF (monthly 2002–2026, annualised): $\sigma_1^2 = 0.02166$, $\sigma_2^2 = 0.00441$, $\sigma_{12} = -0.00098$
  - $(\Sigma \mathrm{w})_1 = 0.5(0.02166) + 0.5(-0.00098) = 0.01034$, $(\Sigma \mathrm{w})_2 = 0.5(-0.00098) + 0.5(0.00441) = 0.00172$
  - $\sigma_p^2 = 0.5(0.01034) + 0.5(0.00172) = 0.00603$, $\sigma_p = 7.76\%$; shares: SPY 85.8%, IEF 14.2%
- Equal weights are not equal risks: volatile assets dominate; risk budgeting chooses w so that the shares equal preset budgets b_i

Equal Risk Contribution

- ERC (Maillard, Roncalli & Teïletche (2010)): $RC_i = RC_j$ for all i, j; computed from a convex problem, then normalised

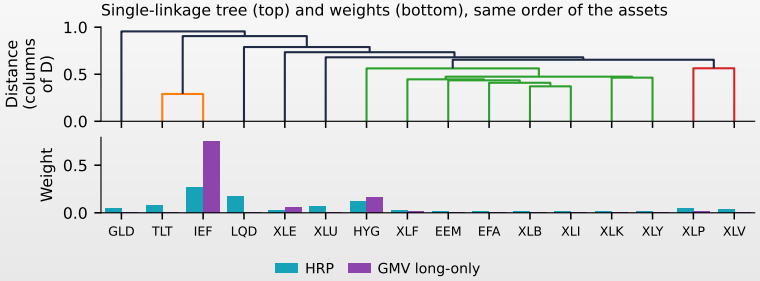
$$\min_{\mathrm{y}>0} \ \tfrac{1}{2}\mathrm{y}^\top \boldsymbol{\Sigma}\mathrm{y} - \tfrac{1}{N}\sum_i \ln y_i, \qquad \mathrm{w} = \mathrm{y}/1^\top\mathrm{y}$$

  - y: unnormalised positive weights; the log term keeps every $y_i > 0$ and its first-order condition gives $y_i(\boldsymbol{\Sigma}\mathrm{y})_i = 1/N$ for all i
- For $\boldsymbol{\Sigma}$ positive definite a solution exists and is unique (strictly convex objective, infinite at the boundary)
- Volatility ordering: $\sigma_{\mathrm{GMV}} \leq \sigma_{\mathrm{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**, $w_1 = \sigma_1^{-1}/(\sigma_1^{-1} + \sigma_2^{-1})$, for any $\rho \in (-1, 1)$
  - SPY (14.7%) and IEF (6.6%): ERC holds 31% SPY, volatility 6.1%, each asset 50% of the variance
- Needs no expected returns, only $\boldsymbol{\Sigma}$; ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

Hierarchical Risk Parity: From a Tree to Weights

- HRP (López de Prado (2016)): three steps, no matrix inversion, all weights positive
  - works even when $\boldsymbol{\Sigma}$ is singular
- **Step 1, tree**
  - correlation distance $d_{ij} = \sqrt{(1-\rho_{ij})/2} \in [0,1]$; ρ_{ij}: correlation of assets i and j
  - Euclidean distance between the **columns** of $\mathrm{D} = (d_{ij})$: $\tilde{d}_{ij} = \sqrt{\sum_k (d_{ki} - d_{kj})^2}$
  - single linkage on $\tilde{d}_{ij}$ (repeatedly merge the two closest clusters) gives a **dendrogram**
- **Step 2, quasi-diagonalisation**
  - order the assets as the leaves of the tree, so that similar ones are adjacent
- **Step 3, recursive bisection**: split the **ordered list** in two halves
  - each half gets weight in inverse proportion to its variance $\tilde{\mathbf{w}}^\top \boldsymbol{\Sigma} \tilde{\mathbf{w}}$
  - $\tilde{\mathbf{w}}$: the inverse-variance weights inside the half
  - repeat inside each half until single assets remain
- Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market
- Textbook: Härdle & Simar (2019), cluster analysis
- A worked four-asset example: Seminar 4, B4

HRP on 16 ETFs



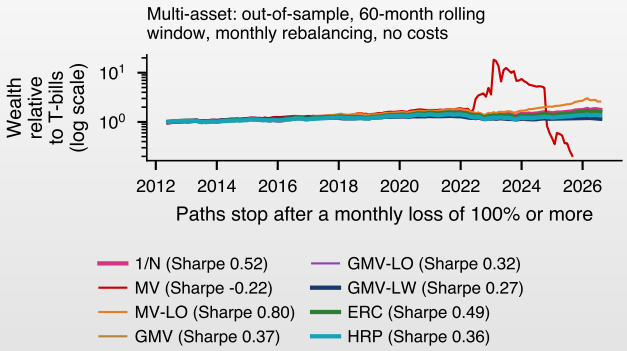
- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026; closest pair TLT and IEF ($\tilde{d} = 0.29$), gold joins last (0.96)
- Equity cluster: cyclical sectors, EFA, EEM and high-yield credit; defensive sectors (utilities, staples, health care) and energy join it late
- HRP: all 16 weights positive, largest IEF 26.7%, LQD 17.6%, HYG 12.6%; long-only GMV: 75.6% in IEF and only five assets above zero

MFM_ch4_hrp

Backtest Design

- Rolling window of 60 months, monthly rebalancing; the weights of month t use months $t-60, \ldots, t-1$ only
- Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- Three universes
  - Sectors: OOS January 2004 – July 2026, 271 months (first window January 1999 – December 2003)
  - Multi-asset and combined: OOS May 2012 – July 2026, 171 months
- Annualised mean, volatility and Sharpe ratio of excess returns; maximum drawdown of total-return wealth
- Every Sharpe ratio difference is reported with a test (next slides), and every test run enters the multiple-testing correction

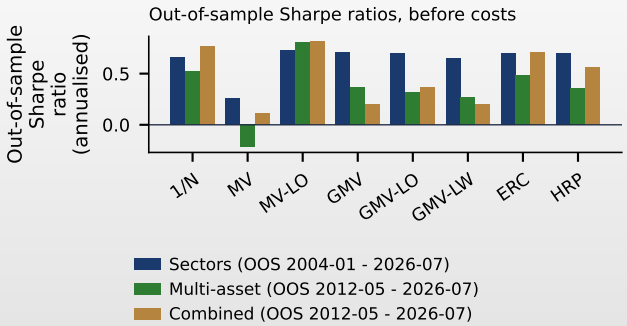
Out-of-Sample Growth: Multi-Asset



- Eight multi-asset ETFs; wealth relative to a Treasury-bill account, before costs
- MV collapses: volatility 518% a year, turnover 44 times wealth per month; a monthly loss above 100% ends its wealth path
- MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- GMV rules: lowest volatility (3.8%–5.4%), low Sharpe (0.27–0.37); they load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT −0.5%, IEF −0.2% a year)
- Is MV-LO really better, or lucky? A test is needed

MFM_ch4_oos_backtest

Sharpe Ratios Across Universes



- 60-month rolling window, monthly rebalancing, annualised, before costs; complete tables for all three universes in the ▸ Appendix
- Sectors: every rule except MV lands within 0.07 of 1/N (0.66); MV 0.26 with drawdown −100%
- The out-of-sample Sharpe of 1/N (0.66, 2004–2026) differs from the simulated one (0.54, 1999–2026) because the periods differ; its match with the true tangency Sharpe (0.66) means nothing
- Combined: 1/N 0.76, MV-LO 0.81, ERC 0.71; the GMV rules 0.20–0.36
- Turnover per month, sectors: 1/N 0.02, ERC 0.03, GMV-LW 0.14, MV-LO 0.23, MV 3.86

MFM_ch4_oos_backtest

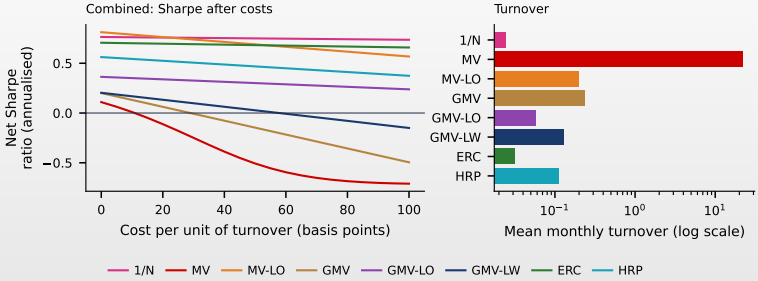
Turnover and Costs: One Worked Month (1/2)

- Between rebalancings weights drift with returns: $w_{i,t}^{+} = w_{i,t}(1 + R_{i,t}) / \sum_j w_{j,t}(1 + R_{j,t})$
  - $R_{i,t}$: **total** return of asset i in month t; the denominator is $1 + R_{p,t}$, one plus the portfolio's total return
- **Turnover**: traded fraction of wealth, $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^{+}|$, summed over all N assets
- Worked month, sectors, 1/N: in January 2004 XLB returned -4.4% and XLK $+4.3\%$, so the weights of 11.11% drifted to 10.50% and 11.45%
  - 1/N total return in January 2004: 1.18% (excess 1.11% plus the Treasury bill 0.07%; Seminar 4, B1)
  - XLB: $\frac{1}{9}(1 - 0.044)/1.0118 = 10.50\%$; XLK: $\frac{1}{9}(1 + 0.043)/1.0118 = 11.45\%$

Turnover and Costs: One Worked Month (2/2)

- Rebalancing back to 1/N in February 2004 trades $TO = 0.019$: 1/N also trades, selling winners and buying losers
  - XLB and XLK contribute $|0.1111 - 0.1050| + |0.1111 - 0.1145| = 0.0061 + 0.0034 = 0.0095$; the other seven sectors about 0.009 together
  - MV-LO, re-estimated on the new window, trades 0.748 in the same month
- Net return with cost c per unit traded: $R^{\text{net}}_{p,t} = R_{p,t} - c\, TO_t$; 1 basis point (bp) $= 0.01\%$, so 20 bp means $c = 0.002$
  - At 20 bp the February trade costs MV-LO $0.002 \times 0.748 = 0.15\%$ of wealth, 1/N $0.002 \times 0.019 = 0.004\%$

Sharpe Ratios after Costs



- Combined universe (16 ETFs); cost c from 0 to 100 bp per unit of turnover; turnover from 0.02 (1/N) to 22 (MV) per month
- Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW; MV-LO overtakes 1/N only for costs below about 22 bp (Seminar 4, B2)
- 1/N and ERC barely move: low turnover is a hidden advantage; report net performance for realistic costs, not only gross

MFM_ch4_costs

Inference on One Sharpe Ratio: Lo (2002)

- Lo (2002), i.i.d. Normal returns: $\sqrt{T}(\widehat{SR} - SR) \to N\left(0,\ 1 + SR^2/2\right)$, with SR monthly, T the number of months: the standard error is $\sqrt{(1 + SR^2/2)/T}$
- 1/N on sectors, $T = 271$: monthly $\widehat{SR} = 0.191$, standard error 0.061
  - Annualised $0.66 \pm 1.96 \times 0.21$: 95% interval $[0.25, 1.08]$
- Annualising by $\sqrt{q}$ ($q = 12$ months per year) is exact only without autocorrelation; in general $SR(q) = \eta(q)\, SR$

$$\eta(q) = \frac{q}{\sqrt{q + 2\sum_{k=1}^{q-1}(q-k)\rho_k}}$$

  - ρ_k: autocorrelation of monthly returns at lag k; positive ρ_k lower η, negative ρ_k raise it; $\rho_k = 0$ gives $\eta = \sqrt{q}$
- For 1/N on sectors: $\rho_1 = -0.03$, $\sum_{k=1}^{11} \rho_k = -0.11$; the formula needs the weighted sum $\sum_{k=1}^{11}(12-k)\rho_k = -0.50$
  - $\eta = 12/\sqrt{12 - 2(0.50)} = 12/\sqrt{11.0} = 3.62$ instead of $\sqrt{12} = 3.46$, annual Sharpe $0.191 \times 3.62 = 0.69$

Testing a Difference in Sharpe Ratios

- H_0: $SR_1 = SR_2$ for two strategies evaluated on the same months; Jobson & Korkie (1981): a test under i.i.d. Normal returns
- Ledoit & Wolf (2008): a robust version, by the delta method

$$\hat{\Delta} = \widehat{SR}_1 - \widehat{SR}_2 = f(\hat{\mathrm{v}}), \qquad f(\mathrm{v}) = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$$

  - r_{1t}, r_{2t}: excess returns of the two strategies; μ_1, μ_2: their means; $\mathrm{v} = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$
  - standard error $\sqrt{\nabla f^\top \hat{\boldsymbol{\Psi}} \nabla f / T}$; ∇f: the gradient of f
  - $\hat{\boldsymbol{\Psi}}$: a HAC (heteroskedasticity and autocorrelation consistent) covariance of the four moments
- Strategies on the same assets are highly correlated: small differences can be significant

The HAC Test Step by Step

- Gradient of f with respect to $\mathrm{v} = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$, writing $\sigma_i^2 = \mathbb{E}[r_i^2] - \mu_i^2$

$$\nabla f = \Big(\frac{\mathbb{E}[r_1^2]}{\sigma_1^3},\ -\frac{\mathbb{E}[r_2^2]}{\sigma_2^3},\ -\frac{\mu_1}{2\sigma_1^3},\ \frac{\mu_2}{2\sigma_2^3} \Big)$$

  - these are derivatives in the **uncentred** second moments; a gradient in (μ_i, σ_i^2) does not fit this $\hat{\Psi}$

- $\hat{\Psi}$: long-run covariance of $\mathrm{y}_t = (r_{1t} - \hat{\mu}_1,\ r_{2t} - \hat{\mu}_2,\ r_{1t}^2 - \widehat{\mathbb{E}r_1^2},\ r_{2t}^2 - \widehat{\mathbb{E}r_2^2})$

- HAC p-values in this chapter's tables: Bartlett (Newey–West) kernel with $\lfloor 4(T/100)^{2/9} \rfloor = 4$ lags for every T from 107 to 271; $z = \hat{\Delta}/\mathrm{SE}$, $p = 2[1 - \Phi(|z|)]$

- The bootstrap studentizes with a different SE (prewhitened quadratic-spectral kernel, Appendix), so the two statistics differ slightly
  - sectors, MV-LO minus 1/N: $\hat{\Delta} = 0.063$; Bartlett SE 0.120, $z = 0.53$, HAC $p = 0.60$; quadratic-spectral (QS) SE 0.127, studentized statistic 0.50

The Block Bootstrap

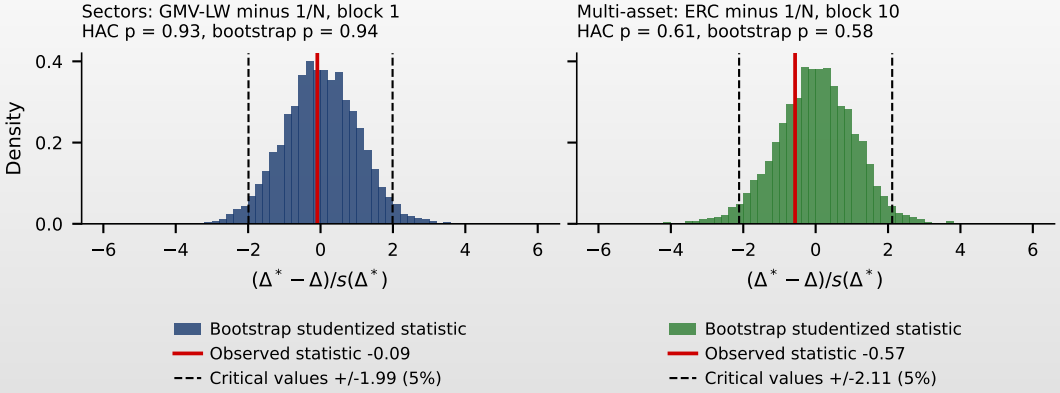


Bradley Efron, inventor of the bootstrap (1979)

Photo: Ryan K. Morris (2007); public domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Ledoit & Wolf (2008), Section 3.2.2: circular blocks of **pairs** (r_{1t}, r_{2t}), wrapping around the end of the sample, so the correlation of the two strategies is kept
- Centre and studentize (divide by an estimated standard error): bootstrap $|\hat{\Delta}^* - \hat{\Delta}|/s(\hat{\Delta}^*)$; asymptotically pivotal (its limit distribution has no unknown parameters), more accurate than the Normal approximation of the HAC test
- $p = \left[\#\{\tilde{d}^{*m} \geq |\hat{\Delta}|/s(\hat{\Delta})\} + 1\right]/(M+1)$, $M = 4999$ draws; $\tilde{d}^{*m}$: the studentized statistic in draw m, $s(\cdot)$: standard error (SE in the seminar), $\#\{\cdot\}$: number of draws
- Block length chosen by the calibration of Ledoit & Wolf (2008), Algorithm 3.1 (▸ Appendix): 1 for six of the seven sector rules; between 1 and 10, most often 10, for multi-asset and combined rules (every block in the 21-test table of the Appendix)

Is Anything Better than 1/N? (1/2)



MFM_ch4_sharpe_test

Is Anything Better than 1/N? (2/2)

- Bootstrap distributions of the studentized difference against 1/N; red: observed statistic; dashed: bootstrap critical values
- Sectors, GMV-LW minus 1/N: -0.02, 95% interval $[-0.36, 0.33]$, HAC $p = 0.93$, bootstrap $p = 0.94$ (block 1); multi-asset, ERC minus 1/N: -0.03, $[-0.15, 0.08]$, HAC $p = 0.61$, bootstrap $p = 0.58$ (block 10): so correlated with 1/N that the interval is narrow, yet it covers zero
- Sectors: no rule differs from 1/N at 5%; the smallest p-value is ERC's ($+0.04$, HAC $p = 0.09$)
- Multi-asset: MV-LO $+0.29$ is **not** significant (HAC $p = 0.12$, bootstrap 0.16)

MFM_ch4_sharpe_test

Many Rules, Many Tests

Universe	Rule minus 1/N	$\hat{\Delta}$	HAC p	Bootstrap p
Multi-asset	MV	-0.74	0.01	0.06
Multi-asset	HRP	-0.16	0.07	0.05
Combined	GMV	-0.56	0.03	0.05
Combined	GMV-LO	-0.40	0.05	0.08
Combined	GMV-LW	-0.56	0.02	0.03
Combined	MV-LO	$+0.05$	0.82	0.82

- No rule beats 1/N significantly in any universe
- Without a multiple-testing adjustment, the GMV rules have a significantly lower Sharpe than 1/N on the combined universe: low risk, but even lower return
- The table shows 6 of the 21 two-sided tests, H_0: $\Delta_k = 0$ (7 rules $\times$ 3 universes; all 21 in the ▸ Appendix). Holm over all 21: smallest adjusted p is 0.29 (HAC, 21×0.014) and 0.68 (bootstrap, 21×0.032): nothing survives (next slide; Chapter 3)
- A different question, the best of the 24 configurations (8 rules $\times$ 3 universes): the deflated Sharpe ratio (DSR) of Bailey & López de Prado (2014) gives $0.84 < 0.95$ for MV-LO on 16 ETFs, but the verdict hinges on the effective number of trials (▸ Appendix)

Holm and Benjamini–Hochberg on This Chapter's Tests

- **FWER** (family-wise error rate): the probability of at least one false rejection in the family; **FDR** (false discovery rate): the expected share of false rejections among the rejections
- Sort the m p-values, $p_{(1)} \leq \cdots \leq p_{(m)}$
  - **Holm** (controls the FWER under any dependence): $\tilde{p}_{(i)} = \max_{j \leq i} \min\{1, (m - j + 1)\, p_{(j)}\}$
  - **Benjamini–Hochberg** (Benjamini & Hochberg (1995); controls the FDR): $\tilde{p}_{(i)} = \min_{j \geq i} \min\{1, \frac{m}{j}\, p_{(j)}\}$
  - reject where $\tilde{p} < 0.05$ (Holm) or $\tilde{p} <$ the chosen FDR (BH)
- The four smallest two-sided HAC p-values, $m = 21$

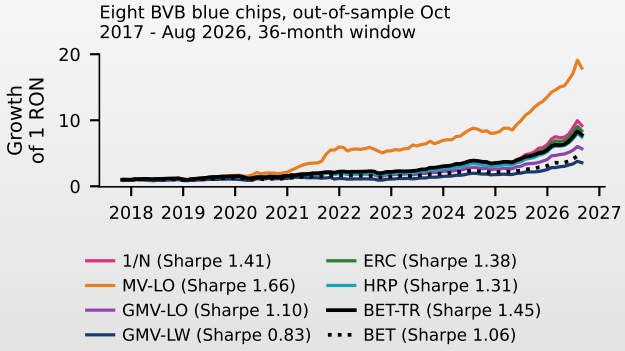
Test	$p_{(j)}$	**Holm**	$\frac{21}{j}\, p_{(j)}$
Multi-asset, MV	0.0138	$21 \times 0.0138 = 0.289$	0.289
Combined, GMV-LW	0.0238	$20 \times 0.0238 = 0.475$	0.249
Combined, GMV	0.0304	$19 \times 0.0304 = 0.577$	0.213
Combined, GMV-LO	0.0477	$18 \times 0.0477 = 0.858$	0.250

- BH takes the minimum from the bottom up: the first three all become 0.213; no rejection at an FDR of 10%

Setup: a Romanian Investor

- Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- Sharpe ratios with $R_f = 0$ (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- Earlier Romanian evidence on diversification: Dragotă, Șerbănescu & Pele (2008)

BVB Portfolios versus BET-TR



- Out of sample, October 2017 – August 2026 (107 months), before costs, $R_f = 0$
- BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.06; the gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- 1/N 1.41, ERC 1.38, HRP 1.31: close to BET-TR
- Rule minus BET-TR: MV-LO 1.66, $+0.21$ (HAC $p = 0.44$, bootstrap 0.53); GMV-LW 0.83, -0.63 (HAC $p = 0.05$, bootstrap 0.10)
- Average correlation of the eight stocks 0.44: little room for diversification inside the BVB; the latest GMV-LO puts 70% in TLV
- Open question: does diversification **beyond** the BVB pay? (Seminar 4, C2)

MFM_ch4_bvb

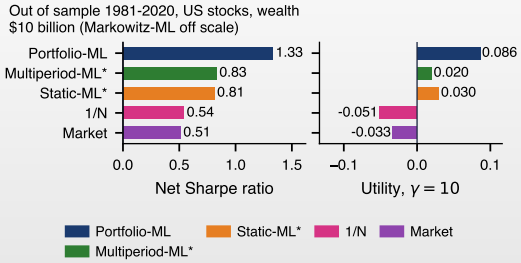
Which Frontier Can a Large Investor Reach? (1/2)

- The central idea: strategies are ranked by the frontier **net of trading costs**
- Jensen, Kelly, Malamud & Pedersen (2026), *Review of Financial Studies*
  - how should portfolios be built from machine-learning (ML) return forecasts when every trade moves prices?
- **Implementable efficient frontier** (eq. 12): for each volatility σ, the highest expected return **net of trading costs**

$$k(\sigma) = \max_{\pi} \; E[r_{t+1}^{\text{net}}] \quad \text{s.t.} \quad E[\pi_t^\top \boldsymbol{\Sigma} \pi_t] = \sigma^2$$

  - π_t: vector of risky-asset weights at t (the paper's notation; not the implied returns $\boldsymbol{\pi}$ of Black–Litterman); r_{t+1}^{net}: portfolio return after trading costs
  - $k(\sigma)$: the best net expected return at volatility σ
- Proposition 1: the net Sharpe ratio $k(\sigma)/\sigma$ falls along the frontier
  - a larger investor faces a lower frontier

The Paper's Results: Net Sharpe and Utility



- Jensen, Kelly, Malamud & Pedersen (2026), Table 2: net of costs, 1981–2020, $\gamma = 10$; utility (eq. 11) = annual net return $-\frac{\gamma}{2}\times$ annual variance
- Markowitz-ML (mean–variance on ML return forecasts, no costs): gross Sharpe 2.06, turnover 79.36, leverage 74.61: net return extremely negative
- Portfolio-ML: net Sharpe 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year)
- Multiperiod-ML* (forecasts for 12 horizons): net Sharpe 0.83, utility 0.020
- Static-ML* (one-period costs; * = extra tuning layer) 0.81, 0.030; 1/N 0.54, −0.051; market 0.51, −0.033: only cost-aware rules have positive utility

MFM_ch4_implementable_frontier

Which Frontier Can a Large Investor Reach? (2/2)

- Cost as a fraction of wealth W_t (eqs. 4, 7), quadratic in the trade

$$c_t = \tfrac{W_t}{2}(\pi_t - g_t\pi_{t-\mathbf{1}})^\top \mathbf{\Lambda}(\pi_t - g_t\pi_{t-\mathbf{1}}), \qquad \mathbf{\Lambda} = \mathrm{diag}(0.2/V_i)$$

  - g_t: drifts last month's weights; V_i: the dollar volume of asset i
- Optimal rule: trade only part of the way towards an **aim portfolio** A_t (Gârleanu & Pedersen (2013))

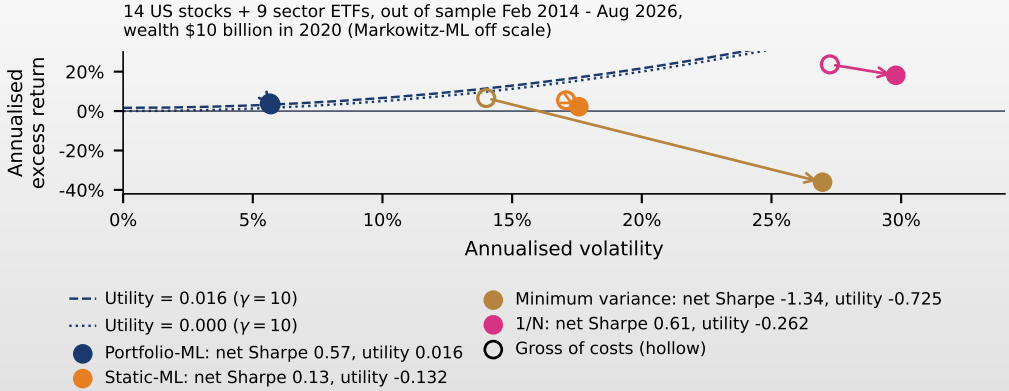
$$\pi_t = m\, g_t\pi_{t-\mathbf{1}} + (\mathbf{I} - m)A_t$$

  - A_t: the portfolio one would hold if trading were free, a weighted average of today's and expected future Markowitz portfolios
  - m: an $N \times N$ matrix of trade rates (eq. 14), so $\mathbf{I} - m$ is a matrix too; in the scalar special case of Gârleanu & Pedersen (2013), m is the share of the old position kept each period
  - Portfolio-ML learns A_t from the signals by a ridge regression (least squares with a penalty on the size of the coefficients) whose objective is utility net of costs

Paper and Course Application: What Is Compared?

	Paper	**Course replication**
Assets	NYSE (New York Stock Exchange) stocks above median size (184–805)	14 US stocks + 9 sector ETFs
Signals	115 characteristics (Jensen, Kelly & Pedersen (2023))	5 price and volume signals
Out of sample	1981–2020, monthly	February 2014 – August 2026 (151 months)
Investor	$\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market	same; wealth scaled by the cumulative SPY return before and after 2020
Utility	annual net return $-\frac{\gamma}{2}\times$ annual variance	same: $12\times$ monthly mean $-\frac{\gamma}{2}\times 12\times$ monthly variance

Gross and Net of Costs on the Course Data (1/2)



MFM_ch4_implementable_frontier

Gross and Net of Costs on the Course Data (2/2)

- Hollow marker: before costs; filled: after the quadratic cost c_t with $\mathbf{\Lambda} = \mathrm{diag}(0.2/V_i)$ (eq. 35 of the paper); dashed curve: pairs with the utility of Portfolio-ML (0.016), dotted: utility 0
- Portfolio-ML barely moves: 3.8% gross, 3.3% net, 5.7% volatility; net Sharpe 0.57
  - utility, annual units: $3.26\% - \frac{10}{2} \times (5.69\%)^2 = 0.0326 - 0.0162 = 0.016$ (net mean and net volatility)
- 1/N keeps 18.1% net (Sharpe 0.61), but at 29.8% volatility its utility is -0.26
- Minimum variance: $+6.6\%$ gross, -36.1% net
- Minimum variance has a leverage of 2.6; rebalancing it costs 43% a year
- Limits: 23 assets, 5 signals, no factor model for $\mathbf{\Sigma}$; the frontier that matters is the one after costs (Seminar 4, C3)

MFM_ch4_implementable_frontier

Open Questions

- Questions
  - does any **estimated** allocation rule beat 1/N out of sample, net of costs?
  - does it still beat 1/N once every rule, window and universe tried is counted?
- Live evidence points both ways
  - Nonlinear shrinkage of Σ improves GMV out of sample in large cross-sections: Ledoit & Wolf (2017)
  - Combining tangency, GMV and 1/N under estimation risk: Kan, Wang & Zhou (2022)
  - Backtests need haircuts for the number of specifications tried: Harvey & Liu (2015)
- Formal statement: the family H_0 of Many Rules, Many Tests against H_1: $\Delta_k > 0$ for some k, with the error rate fixed for the whole family
- One pass of the loop with an AI assistant: literature with DOIs, competing hypotheses, code on the course's monthly data, a pre-registered family of tests, a referee-style critique (prompts in the Appendix)
- Falsified by: a positive $\hat{\Delta}_k$ that survives a family-wise correction, costs and a second sample

What the Researcher Must Check

- **Look-ahead**: weights at t built with month-t returns
  - test: assert that the last estimation date is before t
- **Survivorship and selection**: an ETF list chosen today
  - test: use the universe available at each date, including later delisted ETFs, or disclose the limitation
- **Multiple testing**: 7 rules $\times$ 3 universes already give 21 tests
  - test: Holm (FWER) or Benjamini–Hochberg (FDR) adjusted p-values
- **Power**: $\hat{\Delta}$ has a standard error near 0.18 on 171 months
  - test: compute the detectable effect before running
- **Return type**: portfolios aggregate **simple** returns, Sharpe ratios use **excess** returns
  - test: recompute one month step by step
- **Invented references**: an unresolvable DOI or a title mismatch
  - the claim is dropped and the error goes into `AI_ERRORS.md`

Mini-Case: Data Snooping in This Chapter's Backtest (1/2)

- A typical assistant proposal: "MV-LO beats 1/N on the multi-asset universe"; superiority is one-sided: H_0: $\Delta \leq 0$, $p = 1 - \Phi(\hat{\Delta}/\mathrm{SE})$, Φ the standard Normal distribution function, SE the standard error
- Replicate: $\hat{\Delta} = +0.29$, HAC standard error 0.18, one-sided $p = 0.06$ (two-sided HAC 0.12, bootstrap 0.16)
  - recompute from four decimals, not from the rounded values: $z = 0.2863/0.1836 = 1.56$, $p = 1 - \Phi(1.56) = 0.06$; the rounded $0.29/0.18 = 1.61$ would give 0.054
- Power 0.8 for $\Delta = 0.29$, one-sided: about 435 months (36 years) at 5%, 944 months (79 years) at the Holm level 5%/21
  - months needed: $T_{\mathrm{obs}}\left[(z_{1-\alpha} + z_{0.8})\,\mathrm{SE}/\Delta\right]^2 = 171\left[(1.645 + 0.842) \times 0.1836/0.2863\right]^2 \approx 435$
  - at $\alpha = 0.05/21$, $z_{1-\alpha} = 2.82$, which gives ≈ 944

MFM_ch4_ai_discovery

Mini-Case: Data Snooping in This Chapter's Backtest (2/2)

- Test the family, not the winner: 21 tests (7 rules $\times$ 3 universes; one-sided p-values of all 21 in the ▸ Appendix)
  - Only 7 of 21 differences are positive; the 4 two-sided HAC p-values below 0.05 are all **losses**, which never reject a superiority null
  - One-sided: smallest $p = 0.046$ (sectors, ERC); adjusted Holm 0.96, Benjamini–Hochberg 0.62; 0 discoveries at an FDR of 10%
- Why 0.96 here but 0.29 on Many Rules, Many Tests: a loss has a large one-sided p (about $1 - p_{\text{two-sided}}/2$, e.g. 0.99 for multi-asset MV), so the significant losses drop out; the smallest one-sided p belongs to a gain, 0.046, and Holm multiplies it by 21
- Interpretation: the data cannot rank these rules; the bootstrap interval for MV-LO, $[-0.13, 0.70]$, admits losses and large gains; failure to reject is not proof of equality

MFM_ch4_ai_discovery

Your Turn: A Research Project Idea

- **Questions**
  - does nonlinear shrinkage (Ledoit & Wolf (2017)) beat 1/N on the 16 ETFs of this chapter, after 10–50 bp costs?
  - does a three-fund combination (Kan, Wang & Zhou (2022)) beat 1/N on the same ETFs, after the same costs?
- **Replicate first**: the combined-universe results (1/N Sharpe 0.76, MV-LO 0.81) and the adjusted *p*-values of the mini-case
- **Pre-register**, before any result
  - hypotheses, family of tests, error rate and power
  - a placebo: random long-only weights with the same turnover
- **Report**: code on the course's monthly data, a table of all specifications, a referee-style self-critique
- **Log**: prompts in `AI_USE.md`; every error caught and the test that caught it in `AI_ERRORS.md`

Conclusions: Which Rule, When?

Rule	Objective	Inputs	Main failure mode	Evidence in this chapter
MV	maximum Sharpe	$\hat{\boldsymbol{\mu}}$, $\hat{\boldsymbol{\Sigma}}^{-1}$	error maximisation, explosive weights	worst in every universe; insolvent path
MV-LO	maximum Sharpe, $w_i \geq 0$	$\hat{\boldsymbol{\mu}}$, $\hat{\boldsymbol{\Sigma}}$	corner solutions, turnover	best point estimates, never significant
GMV, GMV-LW	minimum variance	$\hat{\boldsymbol{\Sigma}}$ (shrunk)	loads on low-return assets	lowest risk, loses to 1/N on 16 ETFs (unadjusted)
ERC, HRP	balanced risk	$\hat{\boldsymbol{\Sigma}}$	a tilt of 1/N; HRP trades more	never significantly better than 1/N
Black–Litterman	benchmark plus views	w_b, views, $\boldsymbol{\Omega}$	confidence set too high	moves only the assets in the views
1/N	none	none	ignores risk differences	hard to beat after costs

- Evaluate every rule out of sample, net of costs, with a test of the Sharpe difference and a count of all tests; these verdicts are specific to these samples

Quick Checks with Toy Numbers (1/2)

- **Three rules, two assets**: $\sigma_1 = 30\%$, $\sigma_2 = 15\%$, $\rho = 0.2$
  - compute the weights and volatilities of GMV, ERC and 1/N; check $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
  - what share of the 1/N variance comes from asset 1? (Two Assets; Risk Contributions; Equal Risk Contribution)
- **Optimism of the plug-in Sharpe**: $N = 4$ assets, $T = 120$ months, true $\theta = 0.50$ a year
  - what in-sample maximum Sharpe (root mean square, annual) should you expect? (The Plug-in Sharpe Ratio Is Biased)
- **Noise band**: $N = 10$, $T = 40$, correlation matrix
  - compute the Marchenko–Pastur band; is a sample eigenvalue of 2.0 evidence of structure? (Why the Sample Covariance Matrix Fails)

Quick Checks with Toy Numbers (2/2)

- **One view, one asset**: $\pi = 4\%$, $\tau\sigma^2 = 0.0009$, view 10%
  - posterior mean and view weight for $\Omega = 0.0009$ and for $\Omega = 0.0003$ (Views and the Posterior)
- **One Sharpe ratio**: monthly $\widehat{SR} = 0.15$, $T = 120$, i.i.d. returns from the Normal distribution
  - annual Sharpe and its 95% interval: is it significantly positive? (Inference on One Sharpe Ratio)
- **Four tests**: $p = 0.004,\ 0.020,\ 0.030,\ 0.300$
  - Holm and Benjamini–Hochberg adjusted p-values; how many rejections at 5% under each? (Holm and Benjamini–Hochberg)
- Answers: Appendix

Check Yourself, and Next

Retrieval questions

- Why does the in-sample tangency Sharpe overstate the achievable one? (plug-in Sharpe; Seminar 4, A8)
- By how much does it overstate it for $N = 9$, $T = 60$?
- Why does a long-only constraint act like shrinkage? (Jagannathan–Ma and its two-asset example)
- Why can a small Sharpe difference between ERC and 1/N be nearly significant while a larger one between GMV and 1/N is not? (HAC test: compare the standard errors 0.023 and 0.159 in the Appendix table)

Links to other chapters

- **Chapter 3**: the tangency portfolio and the CAPM; factor covariance models
- **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- **Chapter 13**: machine learning for returns and portfolios; backtest overfitting

The Frontier in Closed Form, Step by Step

- Two constraints, two multipliers: $\mathcal{L} = \frac{1}{2}\mathrm{w}^\top \boldsymbol{\Sigma}\mathrm{w} - \lambda_1(\mathrm{w}^\top \boldsymbol{\mu} - m) - \lambda_2(\mathrm{w}^\top 1 - 1)$
- First-order condition: $\boldsymbol{\Sigma}\mathrm{w} = \lambda_1\boldsymbol{\mu} + \lambda_2 1$, so $\mathrm{w} = \lambda_1\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \lambda_2\boldsymbol{\Sigma}^{-1}1$
- The constraints, with $A = 1^\top\boldsymbol{\Sigma}^{-1}1$, $B = 1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, $C = \boldsymbol{\mu}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, $D = AC - B^2$

$$\boldsymbol{\mu}^\top \mathrm{w} = \lambda_1 C + \lambda_2 B = m, \qquad 1^\top \mathrm{w} = \lambda_1 B + \lambda_2 A = 1$$

- Solving the 2×2 system: $\lambda_1 = (Am - B)/D$, $\lambda_2 = (C - Bm)/D$
- Variance: $\mathrm{w}^\top\boldsymbol{\Sigma}\mathrm{w} = \mathrm{w}^\top(\lambda_1\boldsymbol{\mu} + \lambda_2 1) = \lambda_1 m + \lambda_2 = \dfrac{Am^2 - 2Bm + C}{D}$
- Check: at $m = B/A$ the variance is $(C - B^2/A)/D = 1/A$, the GMV
- Shapes: σ^2 is quadratic in m (a parabola in the (σ^2, m) plane); $\sigma^2 - A(m - B/A)^2/D = 1/A$ is a hyperbola in the (σ, m) plane

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Wishart Moments and the Distribution of Weights

- i.i.d. $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ returns, maximum-likelihood estimators: $\hat{\boldsymbol{\mu}} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/T)$, $T\hat{\boldsymbol{\Sigma}} \sim W_N(T-1, \boldsymbol{\Sigma})$ (Wishart: the matrix analogue of χ^2, $T-1$ degrees of freedom), independent
- Inverse Wishart moment, finite for $T > N+2$: $\mathbb{E}[\hat{\boldsymbol{\Sigma}}^{-1}] = \frac{T}{T-N-2}\boldsymbol{\Sigma}^{-1}$, an **upward** bias that grows with N/T
- Conditioning on $\hat{\boldsymbol{\mu}}$ and using $\mathbb{E}[\mathrm{x}^\top \mathrm{K} \mathrm{x}] = \mathrm{tr}(\mathrm{K}\,\mathrm{Var}(\mathrm{x})) + \mathbb{E}[\mathrm{x}]^\top \mathrm{K}\,\mathbb{E}[\mathrm{x}]$ (any random vector x, fixed matrix K, tr: trace) gives $\mathbb{E}[\hat{\theta}^2]$ of the lecture (Seminar 4, A8)
- GMV weights with Wishart $\hat{\boldsymbol{\Sigma}}$ and T large enough relative to N: $\hat{\mathrm{w}}_{\mathrm{GMV}}$ is unbiased and its exact distribution and covariance are known (Okhrin & Schmid (2006))
- Tangency weights divide by $1^\top \hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\mu}}$, which can be near zero: very heavy tails
- Estimation-risk-optimal mixtures of tangency, GMV and 1/N: Kan & Zhou (2007), Kan, Wang & Zhou (2022)

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Tangency Weights by Regression, Britten-Jones (1999)

- Britten-Jones (1999): regress the constant 1 on the excess returns, **without** a constant term

$$1 = \mathrm{b}^\top \mathrm{r}_t + u_t, \qquad \hat{\mathrm{b}} = (\mathrm{X}^\top \mathrm{X})^{-1} \mathrm{X}^\top 1_T \propto \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$$

- r_t: vector of excess returns in month t; X: $T \times N$ matrix with rows $\mathrm{r}_t^\top$; 1_T: vector of T ones; u_t: regression error
- Sherman–Morrison on $\mathrm{X}^\top \mathrm{X}/T = \hat{\boldsymbol{\Sigma}} + \hat{\boldsymbol{\mu}}\hat{\boldsymbol{\mu}}^\top$ gives the proportionality (Seminar 4, A9); $\hat{\mathrm{b}}/1^\top \hat{\mathrm{b}}$ = sample tangency weights if $1^\top \hat{\mathrm{b}} > 0$
- OLS t-test of $b_i = 0$: asset i has zero weight in the tangency portfolio; F-test of $\mathrm{Rb} = 0$ with $\mathrm{R} = [\mathrm{I}_{N-1}, -1_{N-1}]$: "the tangency portfolio is 1/N"
- Exact finite-sample t and F distributions under i.i.d. Normal returns; HAC standard errors otherwise
- Nine sectors, January 1999 – July 2026 ($T = 331$): normalised $\hat{\mathrm{b}}$ reproduces the tangency weights; largest $|t| = 1.12$ (financials), HAC 1.19
- H_0: tangency = 1/N, eight restrictions: $F(8, 322) = 0.48$, $p = 0.87$; HAC Wald $\chi^2_8 = 4.50$, $p = 0.81$; not rejecting is not proof of equal weights

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The Optimal Shrinkage Intensity

- Ledoit–Wolf choose δ to minimise the expected squared (Frobenius) distance to the true $\boldsymbol{\Sigma}$

$$\delta^* = \max\left\{0, \min\left\{1, \frac{\hat{\kappa}}{T}\right\}\right\}, \qquad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$: sum of the asymptotic variances of the entries of S (noise)
- $\hat{\rho}$: their asymptotic covariances with the target entries
- $\hat{\gamma} = \|\mathsf{F} - \mathsf{S}\|^2$: misspecification of the target (not the risk aversion γ)
- Complete formulas for the constant-correlation target: Ledoit & Wolf (2004b), Appendix; code: the shrinkage Quantlet

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Shrinking the Means and Other Covariance Estimators

- Bayes–Stein (Jorion (1986)): $\hat{\boldsymbol{\mu}}_{\text{BS}} = (1 - \hat{\phi})\,\hat{\boldsymbol{\mu}} + \hat{\phi}\,\hat{\mu}_{\text{GMV}}\mathbf{1}$
  - Empirical-Bayes shrinkage of every mean towards the mean of the GMV portfolio; $\hat{\phi}$ from a Stein argument, larger when N/T is large
- Tu & Zhou (2011): the convex combination of a sophisticated rule and 1/N that minimises the expected out-of-sample loss
- DeMiguel, Garlappi, Nogales & Uppal (2009): norm constraints $\|\mathrm{w}\|_1 \leq \delta$ or $\|\mathrm{w}\|_2 \leq \delta$ nest the long-only GMV, shrinkage GMV and 1/N; the bridge to the lasso portfolios of Chapter 13
- Factor structure plus a sparse residual covariance: POET (principal orthogonal complement thresholding, Fan, Liao & Mincheva (2013)), the link to Chapter 3
- High-dimensional GMV with shrinkage towards a target portfolio: Bodnar, Parolya & Schmid (2018)

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Black–Litterman as Mixed (GLS) Estimation

- Theil & Goldberger (1961): stack the prior and the views as one regression on $\boldsymbol{\mu}$

$$\begin{pmatrix} \boldsymbol{\pi} \\ \mathsf{q} \end{pmatrix} = \begin{pmatrix} \mathsf{I} \\ \mathsf{P} \end{pmatrix} \boldsymbol{\mu} + \begin{pmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{pmatrix}, \qquad \operatorname{Var} \begin{pmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{pmatrix} = \begin{pmatrix} \tau\boldsymbol{\Sigma} & 0 \\ 0 & \boldsymbol{\Omega} \end{pmatrix}$$

- GLS gives exactly $\boldsymbol{\mu}_{\mathsf{BL}}$, with covariance $\mathsf{M} = \left[(\tau\boldsymbol{\Sigma})^{-1} + \mathsf{P}^\top \boldsymbol{\Omega}^{-1} \mathsf{P}\right]^{-1}$
- Equivalent form: $\boldsymbol{\mu}_{\mathsf{BL}} = \boldsymbol{\pi} + \tau\boldsymbol{\Sigma}\mathsf{P}^\top (\mathsf{P}\tau\boldsymbol{\Sigma}\mathsf{P}^\top + \boldsymbol{\Omega})^{-1}(\mathsf{q} - \mathsf{P}\boldsymbol{\pi})$
- $\boldsymbol{\Omega} \to 0$: restricted least squares, $\mathsf{P}\boldsymbol{\mu}_{\mathsf{BL}} = \mathsf{q}$
- Predictive covariance $\boldsymbol{\Sigma} + \mathsf{M}$ in the weights would change all weights; without views it scales them by $1/(1+\tau)$

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Other Risk-Based Rules

- **Inverse volatility**: $w_i \propto 1/\sigma_i$; equals ERC when all correlations are equal
- **Maximum diversification** (Choueifaty & Coignard (2008)): maximise $\mathrm{w}^\top \boldsymbol{\sigma} / \sqrt{\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}}$, the ratio of the average asset volatility to the portfolio volatility; $\boldsymbol{\sigma}$: vector of asset volatilities
- **Minimum variance**: GMV, the extreme risk-based rule
- All ignore expected returns: robust to estimation error, but optimal only under special assumptions

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Results, Sector Universe

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown
1/N	9.4%	14.3%	0.66	0.02	−49%
MV	17.8%	69.0%	0.26	3.86	−100%
MV-LO	10.7%	14.7%	0.72	0.23	−42%
GMV	8.5%	11.9%	0.71	0.23	−33%
GMV-LO	8.1%	11.5%	0.70	0.09	−34%
GMV-LW	7.6%	11.8%	0.65	0.14	−31%
ERC	9.3%	13.2%	0.70	0.03	−47%
HRP	8.9%	12.8%	0.70	0.08	−45%

- January 2004 – July 2026; turnover: mean monthly traded fraction of wealth; drawdown of total-return wealth

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Results, Multi-Asset Universe (8 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown	Sharpe, 50 bp
1/N	4.3%	8.4%	0.52	0.02	−21%	0.50
MV	−113.0%	517.5%	−0.22	44.46	−100%	−0.54
MV-LO	7.1%	8.9%	0.80	0.12	−24%	0.72
GMV	1.4%	3.8%	0.37	0.14	−11%	0.16
GMV-LO	1.7%	5.4%	0.32	0.05	−17%	0.27
GMV-LW	1.1%	4.2%	0.27	0.09	−13%	0.15
ERC	3.6%	7.3%	0.49	0.02	−21%	0.47
HRP	2.4%	6.6%	0.36	0.04	−20%	0.32

- May 2012 – July 2026 (171 months); MV's mean and volatility are arithmetic averages of monthly returns that include a loss above 100%, so only its drawdown has an economic meaning

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Results, Combined Universe (16 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Sharpe, 50 bp
1/N	7.9%	10.3%	0.76	0.02	0.75
MV	28.7%	261.8%	0.11	22.10	−0.51
MV-LO	8.0%	9.8%	0.81	0.20	0.69
GMV	0.8%	4.1%	0.20	0.24	−0.15
GMV-LO	2.0%	5.4%	0.36	0.06	0.30
GMV-LW	0.9%	4.4%	0.20	0.13	0.03
ERC	5.5%	7.8%	0.71	0.03	0.68
HRP	4.0%	7.0%	0.56	0.11	0.47

- May 2012 – July 2026; last column: net of 50 basis points per unit of turnover

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All 21 Tests against 1/N (1/3), Sectors

Rule	$\hat{\Delta}$	HAC SE	p HAC	Holm	p one-sided	p bootstrap	Block
MV	−0.40	0.331	0.22	1.00	0.889	0.30	6
MV-LO	+0.06	0.120	0.60	1.00	0.299	0.63	1
GMV	+0.05	0.159	0.75	1.00	0.377	0.76	1
GMV-LO	+0.04	0.112	0.73	1.00	0.363	0.73	1
GMV-LW	−0.02	0.166	0.93	1.00	0.537	0.94	1
ERC	+0.04	0.023	0.09	1.00	0.046	0.13	1
HRP	+0.04	0.040	0.32	1.00	0.162	0.36	1

- January 2004 – July 2026, $T = 271$; $\hat{\Delta}$: annualised Sharpe of the rule minus that of 1/N, before costs
- HAC: Bartlett, 4 lags, two-sided; Holm over all 21 two-sided HAC tests; one-sided: $1 - \Phi(\hat{\Delta}/\mathrm{SE})$; bootstrap: studentized circular block bootstrap, two-sided, block from Algorithm 3.1

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All 21 Tests against 1/N (2/3), Multi-Asset

Rule	$\hat{\Delta}$	HAC SE	p HAC	Holm	p one-sided	p bootstrap	Block
MV	−0.74	0.300	0.01	0.29	0.993	0.06	10
MV-LO	+0.29	0.184	0.12	1.00	0.059	0.16	10
GMV	−0.15	0.226	0.51	1.00	0.745	0.54	1
GMV-LO	−0.20	0.137	0.14	1.00	0.930	0.15	8
GMV-LW	−0.25	0.191	0.19	1.00	0.903	0.20	4
ERC	−0.03	0.060	0.61	1.00	0.694	0.58	10
HRP	−0.16	0.089	0.07	1.00	0.964	0.05	6

- May 2012 – July 2026, $T = 171$; same columns as for the sectors

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All 21 Tests against 1/N (3/3), Combined

Rule	$\hat{\Delta}$	HAC SE	p HAC	Holm	p one-sided	p bootstrap	Block
MV	−0.66	0.355	0.06	1.00	0.968	0.22	10
MV-LO	+0.05	0.210	0.82	1.00	0.409	0.82	10
GMV	−0.56	0.261	0.03	0.58	0.985	0.05	1
GMV-LO	−0.40	0.203	0.05	0.86	0.976	0.08	6
GMV-LW	−0.56	0.249	0.02	0.48	0.988	0.03	2
ERC	−0.06	0.114	0.61	1.00	0.695	0.62	10
HRP	−0.20	0.152	0.18	1.00	0.908	0.20	8

- May 2012 – July 2026, $T = 171$; 7 of the 21 $\hat{\Delta}$ are positive, and every two-sided HAC $p < 0.05$ is a loss
- One-sided family: Holm $21 \times 0.046 = 0.96$; Benjamini–Hochberg $\min\{21 \times 0.046,\ \frac{21}{2} \times 0.059\} = 0.62$ for the two smallest

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Bootstrap Details and the Block Length

- Studentization: $s(\hat{\Delta}^*)$ from the block sums of each resample; $s(\hat{\Delta})$ on the data from a prewhitened quadratic-spectral kernel (Andrews & Monahan (1992)) with automatic bandwidth (Andrews (1991))
  - the HAC p-values in the tables use the simpler Bartlett kernel with 4 lags (HAC test); the two SEs differ mostly in the second decimal
- Ledoit & Wolf (2008), Algorithm 3.1: choose the block b whose bootstrap interval has actual coverage closest to 95%
  - Fit a first-order vector autoregression to the pair of returns; resample its residuals by a stationary bootstrap (Politis & Romano (1994)), mean block 5
  - Generate $K = 5000$ pseudo-samples of length T; in each, build the 95% interval for $b \in \{1, 2, 4, 6, 8, 10\}$
  - $\hat{g}(b)$: share of intervals that contain the pseudo-true $\hat{\Delta}$; pick $\tilde{b} = \arg\min_b |\hat{g}(b) - 0.95|$
- The grid and K are those of the paper (Sections 4–5); applied to each of the 21 pairs of this chapter
- A non-studentized bootstrap with a fixed 6-month block is not this test: its block is not calibrated for coverage

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The Deflated Sharpe Ratio

- Bailey & López de Prado (2014): the best of N Sharpe ratios is biased upwards even if every true Sharpe ratio is zero
- Expected maximum under H_0: all true Sharpe ratios zero, N independent trials, monthly units as T; on this slide N counts trials (configurations), not assets
  - $\gamma_E = 0.5772$: Euler's constant; $\hat{V}$: the variance of the N estimates
  - Φ: standard Normal distribution function; e: Euler's number

$$SR_0 = \sqrt{\hat{V}}\left[(1-\gamma_E)\,\Phi^{-1}\left(1-\tfrac{1}{N}\right) + \gamma_E\,\Phi^{-1}\left(1-\tfrac{1}{Ne}\right)\right]$$

- Deflated Sharpe ratio: the probability that the true Sharpe of the winner exceeds SR_0, with skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$ of the winner

$$\text{DSR} = \Phi\left(\frac{(\widehat{SR} - SR_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right)$$

- This chapter: 24 configurations (8 rules $\times$ 3 universes); winner MV-LO on 16 ETFs, 0.81 a year
  - As 24 independent trials: $SR_0 = 0.52$ a year, DSR $= 0.84 < 0.95$
  - The configurations are strongly correlated (up to 0.99), so fewer are independent: $N = 8$ gives DSR 0.93, $N = 3$ gives 0.98
- Holm adjusts the whole family of pairwise tests against 1/N; the DSR asks whether the single best configuration is better than luck

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Beyond Equities, Digital Assets

- Petukhina, Trimborn, Härdle & Elendner (2021): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- Pele et al. (2023): are cryptos becoming alternative assets?
- Bitcoin: annualised volatility 172% (monthly, 2010–2026, one surviving coin chosen ex post), IEF 6.6%
- Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically (Seminar 4, B8)
- Robust mean–variance for crypto-assets: Pele (2023)

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Case Study, Method in Detail (1/2)

- **Cost** (eqs. 4, 7)
  - π_t: risky-asset weights; $1 - 1^\top \pi_t$: the weight of the risk-free asset
  - $g_t = \mathrm{diag}\big((1 + r_t^f + r_t)/(1 + g_t^w)\big)$ (eq. 6); r_t^f: risk-free rate, r_t: asset returns, g_t^w: growth of wealth
  - $\mathbf{\Lambda} = \mathrm{diag}(0.2/V_i)$; V_i: average daily dollar volume over 6 months; trading 1% of V_i moves the price by 0.1% (eq. 35, Frazzini, Israel & Moskowitz (2018))
- **Optimal rule** (Proposition 3): $\pi_t = m\, g_t \pi_{t-\mathbf{1}} + (\mathrm{I} - m) A_t$, generalising Gârleanu & Pedersen (2013)
  - A_t: a weighted sum of today's and expected future Markowitz portfolios; m solves eq. (14)
- **Portfolio-ML** (Proposition 4, eq. 40): $A_t = \mathrm{diag}(1/\sigma_{i,t})\, f(s_t)\beta$
  - $\sigma_{i,t}$: volatility of asset i; f: p random Fourier features of the signals s_t
  - $\hat{\beta} = (E_T[\tilde{\mathbf{\Sigma}}_t] + \lambda \mathrm{I})^{-\mathbf{1}} E_T[\tilde{r}_{t+\mathbf{1}}]$: a ridge regression ($\lambda$: penalty) that maximises utility net of costs in closed form (eqs. 24–26)
  - it learns the weights directly, as Brandt, Santa-Clara & Valkanov (2009) do without costs

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Case Study, Method in Detail (2/2)

- **Data** (Section 5.1): NYSE common stocks above the median market capitalisation (184–805 stocks)
  - 115 characteristics of Jensen, Kelly & Pedersen (2023), ranked into $[0, 1]$
  - December 1952 – December 2020, out of sample 1981–2020, monthly
- **Covariance $\boldsymbol{\Sigma}$**: 13 characteristic themes and 12 industries as factors
  - exponential weights with half-lives of 378 days (correlations) and 126 days (variances)
  - investor $\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market (Section 5.1.3)
- **Tuning** (Table 1), chosen each year by the utility of validation backtests since 1971
  - $\lambda \in \{0, e^{-\mathbf{10}}, e^{-\mathbf{9.8}}, \ldots, e^{\mathbf{10}}\}$, $p \in \{2^{\mathbf{6}}, \ldots, 2^{\mathbf{9}}\}$, $\eta \in \{e^{-\mathbf{3}}, e^{-\mathbf{2}}\}$; η: standard deviation of the random weights
- **Rivals** ("*" marks an extra tuning layer, Section 5.3)
  - Markowitz-ML (eq. 32), Static-ML (one-period cost, eq. 34), Multiperiod-ML (forecasts for 12 horizons, eq. 22)
  - factor sorts, 1/N, minimum variance, market

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Case Study, Complete Results

- **Table 2**, Portfolio-ML: turnover 0.25 a month (net Sharpe ratio 1.33)
- **Table 2**, Multiperiod-ML*: net Sharpe 0.83, utility 0.020
  - Portfolio-ML needs no second tuning layer
- **Table 3**: posterior probability that Portfolio-ML has the higher utility
  - 99.9% against Static-ML* and Multiperiod-ML*, 100% against Markowitz-ML
- **Figure 1(B)** (Section 6.2), $\gamma = 10$: size is a cost
  - with \$10 billion, net excess return 14.6% at 11% volatility; with \$1 billion, 16.3%
- **Figure 5**: net of costs, quality is the most important theme; short-term reversal is almost irrelevant
  - monthly autocorrelation -0.04 for 1-month reversal, 0.94 for book-to-market (Section 6.3)
  - persistent signals are cheap to trade
- **Limits stated by the authors**
  - fees and other frictions; computing power and ML methods not available in real time early in the sample (Section 6.1)
  - costs and risk treated as constant over the horizon of each optimisation (Section 5.1.3)
  - among the smallest 20% of stocks, Portfolio-ML ranks behind Multiperiod-ML* and Static-ML* (Figure 8)

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Case Study, the Course Replication

- **Course data**: 14 US stocks and 9 sector ETFs
  - five signals: 1-month log return, log momentum from $t-6$ and from $t-12$ to $t-1$, 1-year volatility, 6-month dollar volume (formulas in Seminar 4, C3)
  - same cost rule, γ, wealth and grid; $\mathbf{\Sigma}$ with the same weights on the 23 assets
  - validation from 2011; out of sample February 2014 – August 2026 (151 months)
- **Portfolio-ML**: cost 0.6% a year, leverage 0.32
- **Other rules**: utility of Static-ML -0.13; minimum variance -0.72; Markowitz-ML extremely negative (leverage 18)
- **Posterior probability** of a higher utility for Portfolio-ML
  - 99.5% against Static-ML, 99.9% against minimum variance, 93.3% against 1/N
- Same ranking of cost-aware against cost-agnostic rules as Table 2
  - Portfolio-ML wins with small, slowly traded positions, not with a higher gross Sharpe ratio (0.68 against 0.87 for 1/N)
- **Limits**
  - 23 assets instead of 184–805; five signals instead of 115
  - no factor model for $\mathbf{\Sigma}$, whose window reaches 10 years only in August 2014 (common prices from August 2004)
  - return forecasts with $p \geq 2^6$ and three validation years instead of ten; no starred rivals
- **Open extension** (Seminar 4, C3)
  - implementable frontiers for wealth of \$1, 10 and 100 billion and $\gamma \in \{1, 5, 10, 20, 100\}$
  - 1/N's trading costs computed at each wealth level

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The Discovery Loop, Reusable Prompts

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature**: *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with 1/N out of sample; for each give DOI, N, T, cost assumption and test."* Resolve every DOI on Crossref and open the table cited
- **Hypotheses**: *"Give three competing explanations for a Sharpe gain over 1/N (better Σ, a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code**: *"Write Python that reads the course's daily price files, aligns prices on common dates and runs a 60-month rolling GMV-LW and 1/N; weights at t may use data up to $t-1$ only."* Replicate first: 1/N 0.76, GMV-LW 0.20 on the combined universe
- **Robustness**: *"Draft a pre-registration: the family of tests (rules $\times$ universes $\times$ windows $\times$ costs), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
- **Critique**: *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report**: every specification tried, adjusted p-values, the power of the design, code and seeds; any specification chosen after seeing results is exploratory

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Answers to the Quick Checks

- **Three rules**: $s_{11} = 0.09$, $s_{22} = 0.0225$, $s_{12} = 0.2 \times 0.30 \times 0.15 = 0.009$
  - GMV: $w_1 = (0.0225 - 0.009)/(0.09 + 0.0225 - 0.018) = 1/7$, volatility 14.3%; ERC (inverse volatility): $w_1 = 1/3$, volatility 15.5%; 1/N: 18.1%
  - 1/N: $w_1(\Sigma w)_1 = 0.5 \times 0.0495 = 0.02475$ of $\sigma_p^2 = 0.0326$: asset 1 carries 76% of the risk with 50% of the weight
- **Plug-in Sharpe**: monthly $\theta^2 = 0.25/12 = 0.0208$; $(120 \times 0.0208 + 4)/(120 - 6) = 0.0570$; $\sqrt{12 \times 0.0570} = 0.83$ a year, against a true 0.50
- **Noise band**: $c = 10/40 = 0.25$, band $[(1 - 0.5)^2, (1 + 0.5)^2] = [0.25, 2.25]$; 2.0 lies inside, so this comparison cannot separate it from noise
- **One view**: equal precisions give view weight $0.0009/(0.0009 + 0.0009) = 0.5$, posterior 7%; with $\Omega = 0.0003$, weight $0.0009/0.0012 = 0.75$, posterior $4\% + 0.75 \times 6\% = 8.5\%$
- **One Sharpe ratio**: SE $= \sqrt{(1 + 0.15^2/2)/120} = 0.092$; annual $0.52 \pm 1.96 \times 0.32$, interval $[-0.10, 1.14]$: not significant
- **Four tests**: Holm $0.016, 0.060, 0.060, 0.300$: one rejection; Benjamini–Hochberg $0.016, 0.040, 0.040, 0.300$: three rejections

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Courses, Code and Data

Advanced courses on quantinar.com

- Hierarchical Clustering
- Minimum Spanning Tree
- Multivariate Statistical Analysis

Code and data

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_04
- Risk-free rate: Kenneth French Data Library

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Acronyms used in this material

- **AI**: Artificial Intelligence
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAPM**: Capital Asset Pricing Model
- **DSR**: Deflated Sharpe Ratio
- **ERC**: Equal Risk Contribution portfolio
- **ETF**: Exchange-Traded Fund
- **FDR**: False Discovery Rate
- **FWER**: Family-Wise Error Rate
- **GLS**: Generalised Least Squares
- **GMV**: Global Minimum-Variance portfolio
- **GMV-LO**: Global Minimum-Variance portfolio, Long-Only
- **GMV-LW**: Global Minimum-Variance portfolio with Ledoit–Wolf covariance
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HRP**: Hierarchical Risk Parity
- **HYG**: iShares iBoxx High Yield Corporate Bond ETF
- **IEF**: iShares 7–10 Year Treasury Bond ETF
- **LO**: Long-Only, no short sales
- **LQD**: iShares iBoxx Investment Grade Corporate Bond ETF
- **ML**: Machine Learning
- **MV**: Mean–Variance tangency portfolio estimated from sample moments
- **MV-LO**: Mean–Variance tangency portfolio, Long-Only
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **OMV**: Österreichische Mineralölverwaltung — *Austrian Mineral Oil Administration (OMV group)*
- **OOS**: Out-Of-Sample
- **POET**: Principal Orthogonal complEment Thresholding
- **US**: United States